Solution

CADET _____ SECTION _____TIME OF DEPARTURE _____

DEPARTMENT OF CHEMISTRY & LIFE SCIENCE

QUIZ 3 – CH402 2025-2026
10 Minutes, 24 Points
25 February 2026

TEXT: Peters, Timmerhaus, & West
SCOPE: Present and future values

References Permitted: FE Reference Manual online.

INSTRUCTIONS

1. Do not mark this quiz until "begin work" is given. You will have 10 minutes.
2. Solve the problems in the space provided. Show all work to receive credit.
3. There are 7 problems on 3 pages in this quiz, not including the cover page.
4. Write your name on the top of each sheet.
5. Show work to receive partial credit.

(TOTAL WEIGHT: 24 POINTS)

DO NOT WRITE IN THIS SPACE

PROBLEM	VALUE	CUT
1	3	a
2	3	c
3	4	d
4	4	c
5	4	d
6	3	b
7	3	c
CUT		
GRADE	24	

Cadet: Solution

The data below pertains to Problems 1 and 2 and shows the estimated before-tax annual cash flows for a project.

Year End	1	2	3	4
Cash Flow	$1,000	$1,000	$1,000	$1,000

1. (3 points) At 6% interest per year compounded annually, the *present* value of the project is most nearly

(A) $3,465
(B) $2,165
(C) $2,700
(D) $4,000

To P given A $\Rightarrow$ (P/A, 6%, 4) = 3.4651

1000 · 3.4651 = $3465

ANS A

2. (3 points) At 6% per year compounded annually, the *present* value of the project with 24% annual taxes is most nearly

A) $1,740
B) $3,040
C) $2,633
D) $1,333

$1000 \cdot (1-.24) \cdot \underbrace{3.4651}_{\text{from 1}} = \$2633$

ANS C

Cadet: Solution

3. (4 points) The present value of the profits from a process is $1,606,000. Assuming a facility life of 18 years, what is the equivalent annual cash flow (A) from the process if the interest rate is 2% compounded annually?

A) $125,240
B) $150,300
C) $200,110
D) $107,120

to A given P ⇒ (A/P, 2%, 18) = .0667

1,606,000 · .0667 = $107,120

ANS D

4. (4 points) A heat exchanger originally cost $21,000 and has a service life of three years with no scrap value. Assuming the cost of the original heat exchanger remains the same (no inflation), how much money must be set aside annually to replace the heat exchanger after 3 years of service at an annual interest rate of 6% compounded annually?

A) $ 7,992
B) $ 5,593
C) $ 6,596
D) $ 6,998

to A given F ⇒ (A/F, 6%, 3) = .3141

21,000 · .3141 = $6596

ANS C

Cadet: Solution

5. (4 points) The annual maintenance costs for a facility are $0 for the first year and increase by $300 each year thereafter (assumed payable at the end of each year). Assuming a facility life of 15 years, what is the present worth of the maintenance costs over the lifetime of the facility if the interest rate is 8% compounded annually?

A) $ 4,510
B) $22,315
C) $29,220
D) $14,366

to P given G → (P/G, 8%, 15) = 47.8857

300 · 47.8857 = $14,366

ANS D

6. (3 points) The power scaling factor for acetic acid plants using the CH_3OH-CO catalytic process is 0.68. If the fixed capital investment for this type of plant producing 9,000 metric tons per year is $8,000,000, the price for a plant producing 18,000 metric tons per year is most nearly

A) $12,120,000
B) $12,820,000
C) $16,000,000
D) $13,700,000

exp. scaling method:

$$8{,}000{,}000 \cdot \left(\frac{18000}{9000}\right)^{.68} = \$12{,}817{,}000$$

ANS B

7. (3 points) The power scaling factor for chlorine plants using electrolysis of NaCl is 0.45. The chemical engineering plant cost index was 394.1 in 2000 and 782.5 in 2025. If the fixed capital investment for this type of plant producing 45,000 metric tons per year was $33,000,000 in the year 2000, the price for a plant producing 90,000 metric tons per year in 2025 was most nearly

A) $66,000,000
B) $23,000,000
C) $91,000,000
D) $45,000,000

$$\left(\frac{90000}{45000}\right)^{.45} \cdot 33{,}000{,}000 = 45{,}000{,}000$$

$$45{,}000{,}000 \cdot \frac{782.5}{394.1} = \$91{,}000{,}000$$

ANS C